

EXP 28: Prolog Program for Medical Diagnosis

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Aim:

To develop a Prolog program that performs a simple medical diagnosis by matching a patient's symptoms against known disease rules.

Algorithm:

- Step 1: Start the program.
- Step 2: Collect the patient's symptoms.
- Step 3: Compare the symptoms with predefined disease rules.
- Step 4: Check whether the symptoms match those of a cold.
- Step 5: If not, check for flu symptoms.
- Step 6: If not, check for malaria symptoms.
- Step 7: Identify the disease whose symptoms match.
- Step 8: Display the diagnosis result.
- Step 9: Stop the program.

Source Code:

```
% Program for Medical Diagnosis

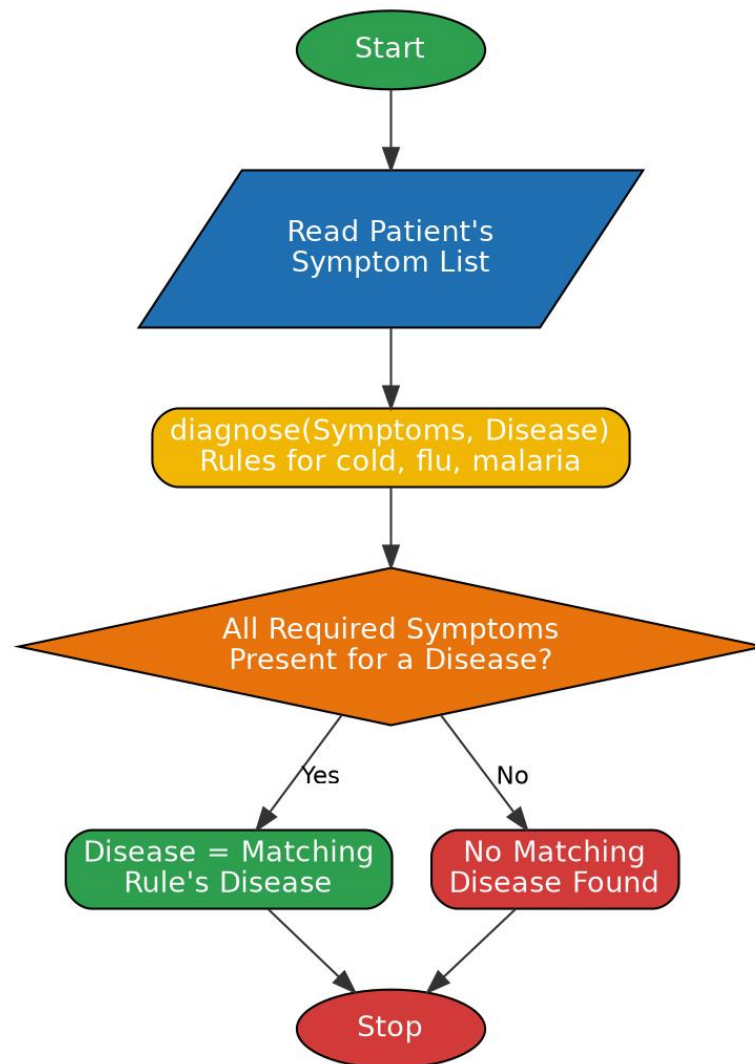
has_symptom(Symptoms, S) :- member(S, Symptoms).

diagnose(Symptoms, cold) :-
    has_symptom(Symptoms, cough),
    has_symptom(Symptoms, sneezing),
    has_symptom(Symptoms, runny_nose).

diagnose(Symptoms, flu) :-
    has_symptom(Symptoms, fever),
    has_symptom(Symptoms, body_ache),
    has_symptom(Symptoms, fatigue).

diagnose(Symptoms, malaria) :-
    has_symptom(Symptoms, fever),
    has_symptom(Symptoms, chills),
    has_symptom(Symptoms, sweating).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
?- cd('C:/Users/yanar/Documents/CSA1717').
true.
```

```
?- [medical].
true.
```

```
?- diagnose([cough, sneezing, runny_nose], Disease).
Disease = cold ,
```

```
?- diagnose([fever, body_ache, fatigue], Disease).
Disease = flu ,
```

```
?- diagnose([fever, chills, sweating], Disease).
Disease = malaria ,
```

Result:

Thus the Prolog program for medical diagnosis was written, executed successfully and the output was verified.